

TEACHING OPEN SCIENCE AND REPRODUCIBILITY

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WHY BOTHER?

We would like our results to be as fully reproducible as possible:

A. Reproducibility is one of the pillars of science

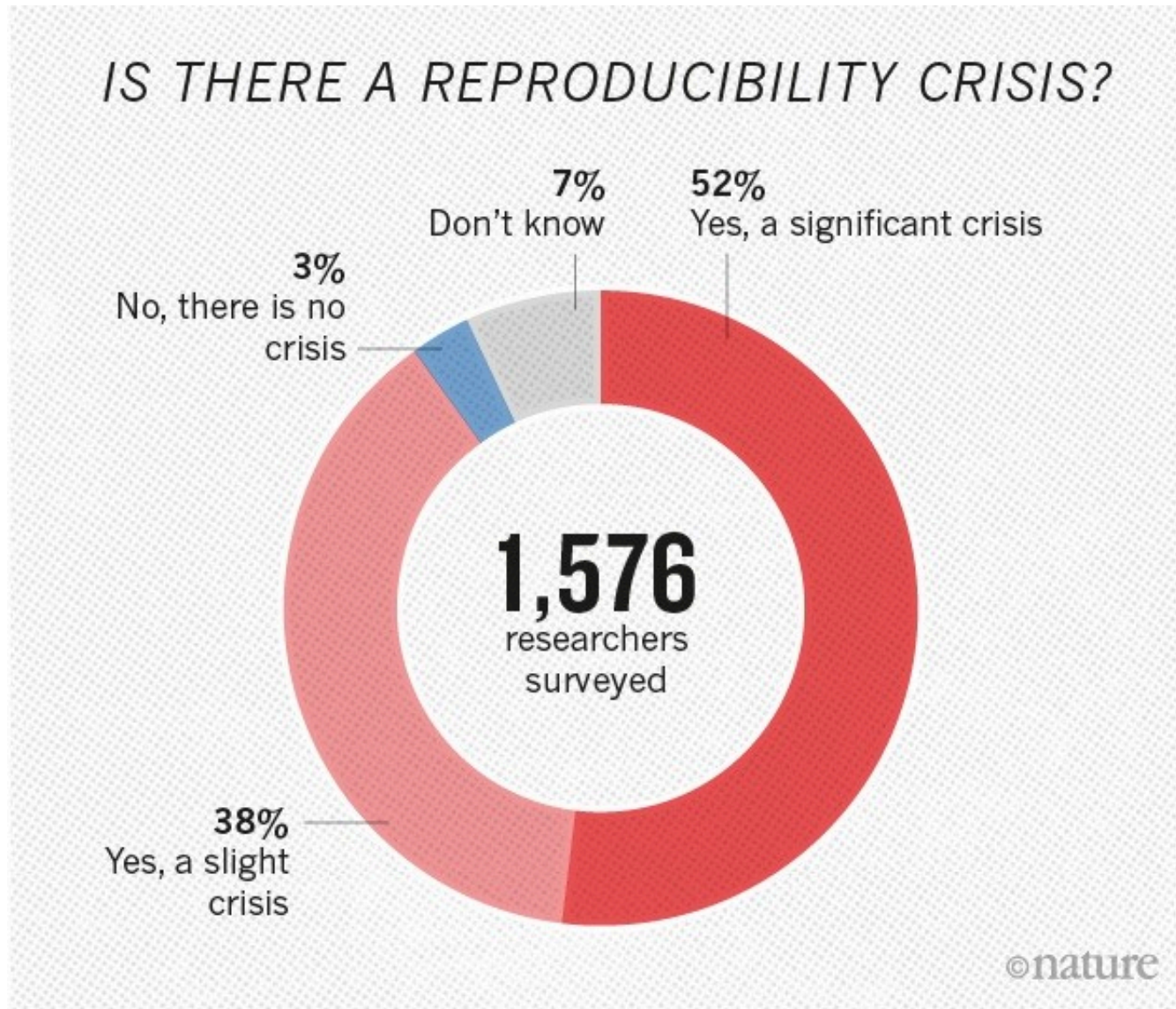
- It is the standard by which to judge scientific claims
- It helps the cumulative growth of knowledge - no duplication of effort

B. Reproducibility may greatly benefit you

- You'll develop better work habits
- Better teamwork - especially with new team members
- Changing or amending your work is much easier
- Higher research impact - more likely to be picked up and cited



CRISIS?



DEFINITION

		Data	
		Same	Different
Analysis	Same	Reproducible	Replicable
	Different	Robust	Generalisable

A result is **reproducible** when the *same* analysis steps performed on the *same* dataset consistently produces the *same* answer.



DEFINITION

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Research results are **replicable** if there is sufficient information available for independent researchers to make the same findings using the same procedures.



TRUE OR FALSE?

In computational sciences - such as statistics - simply having the data and code means that the results are not only replicable, but fully reproducible.

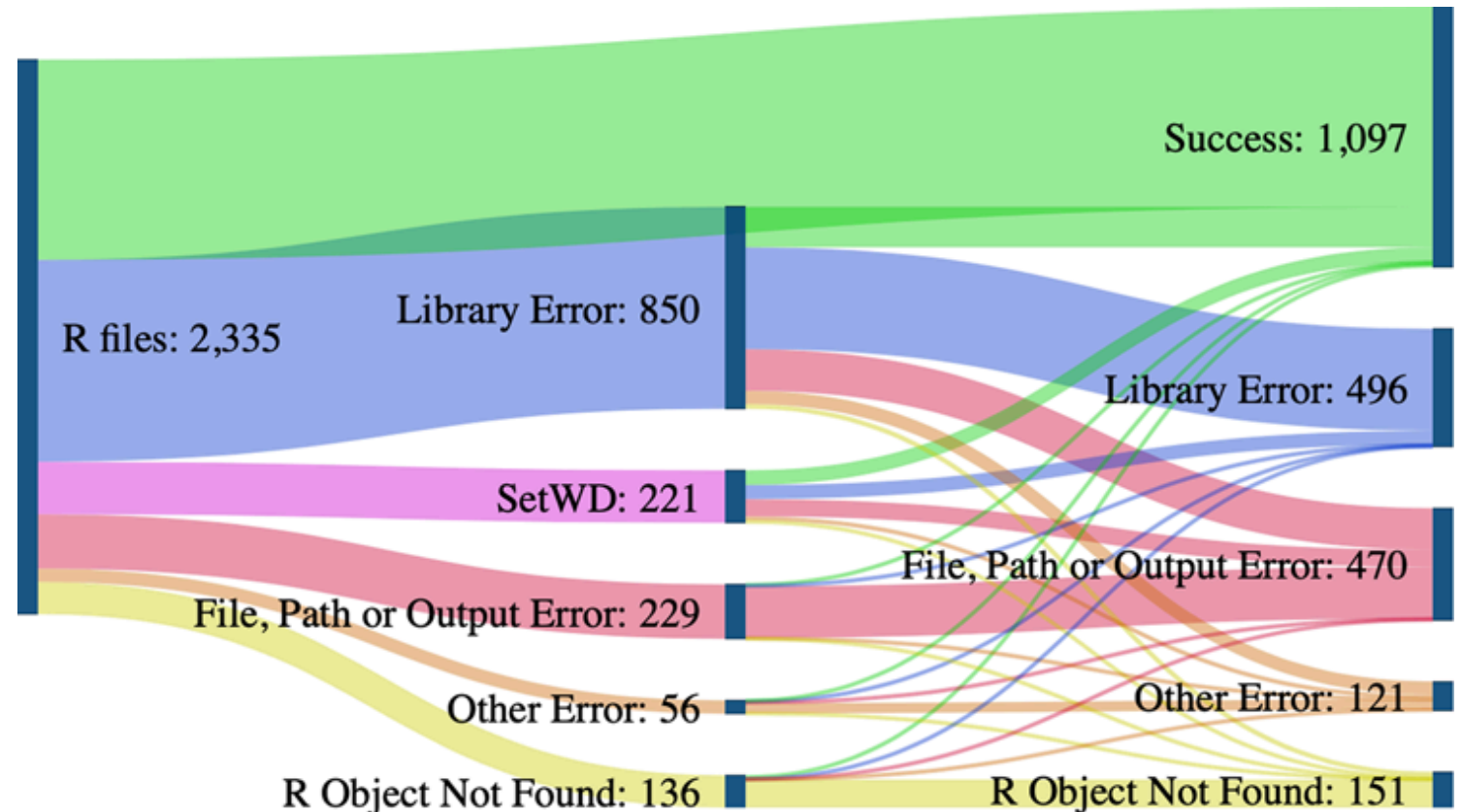
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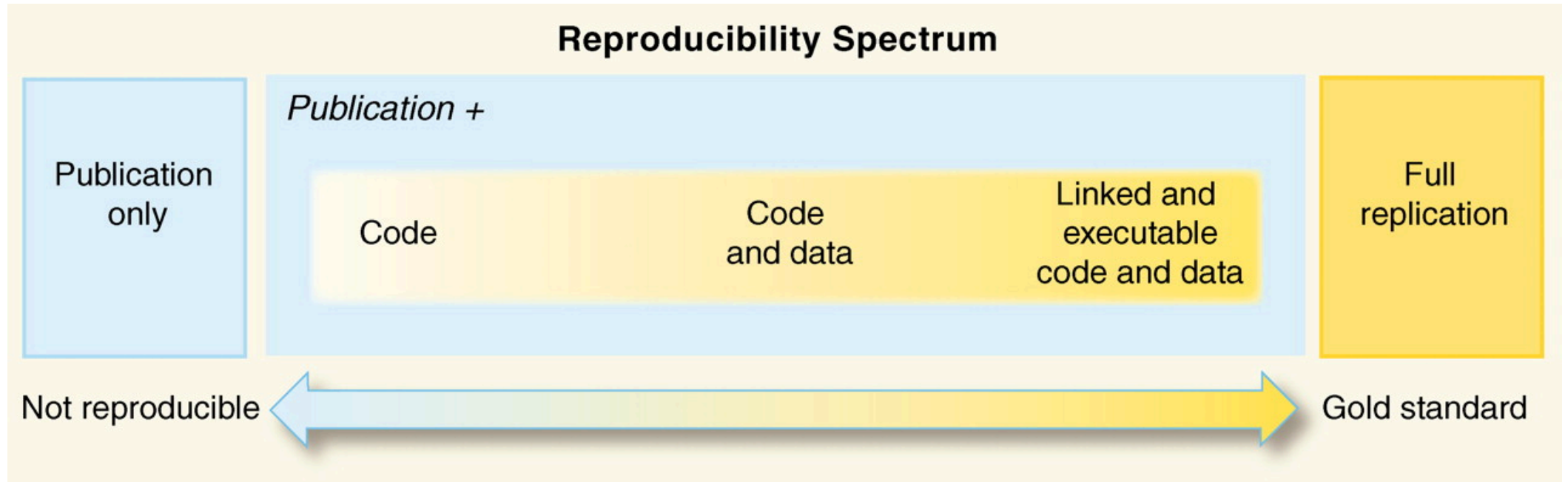
REPRODUCIBILITY OF R SCRIPTS

Reproducible research is not the norm:

74% of R files failed to complete without error



REPRODUCIBILITY SPECTRUM



MAKING RESEARCH REPRODUCIBLE



REPRODUCIBLE BUT NOT REPLICABLE

```
1 set.seed(1)
2 run_simulation()
3 set.seed(2)
4 run_simulation()
5 set.seed(3)
```



TEACHING REPRODUCIBILITY



Utrecht University

Markup Languages and Reproducible Programming in Statistics



COURSE AIMS

Course aims include the development of a publication-ready reproducible research compendium that contains:

- a typeset manuscript following a markup language,
- data and code,
- everything that allows for successful reproduction and reuse of the materials (e.g. a license).

In our course, students are taught various tools and languages, such as Quarto markdown, version control with git, and reproducible environments for R with [renv](#).



FULL COURSE AIMS

- a. Students develop fundamental knowledge and understanding in the state of the art in statistical markup languages and reproducible programming and development
- b. They can determine the most effective markup strategies to address a typesetting problem
- c. They can efficiently organize a reproducible programming and development process
- d. They can produce repositories up to the standards of international programming and coding conventions and initiatives
- e. They can produce publications up to the typesetting standards of international peer-reviewed journals



MARKUP LANGUAGES

L^AT_EX



VERSION CONTROL



LICENSING



SHARING R CODE



RESEARCH COMPENDIUMS



COURSE WEEKS

1. Markup languages
2. Quarto markdown
3. Version control (with git and GitHub)
4. Reproducible research in statistics
5. Developer portfolios
6. Re-usable R code (with R packages and Shiny)



REUSABLE COURSE ELEMENTS



MISSING ELEMENT(S)?





Teaching Reproducibility to Research Master's Students

What resources to use? And which tools/elements are domain agnostic?

HI Oberman

Our course

We teach 'Markup Languages and Reproducible Programming in Statistics' in the second year of the Master's programme 'Methodology and Statistics for the Behavioural, Biomedical and Social Sciences' at Utrecht University (UU).^{1,2}

Course aims include the development of a publication-ready **reproducible research compendium** that contains:

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In our course, students are taught various tools and languages, such as Quarto markdown, version control with *git*, and reproducible environments for R with *renv*.

Reusable course elements

Our course materials draw from resources about the tools and languages that we teach, e.g. software manuals. In principle, these materials are reusable by others who teach reproducibility. But, of course, this is heavily reliant on the specific software framework: in our case, the statistical programming language R.

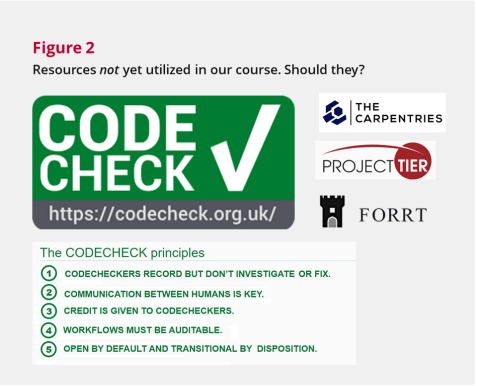
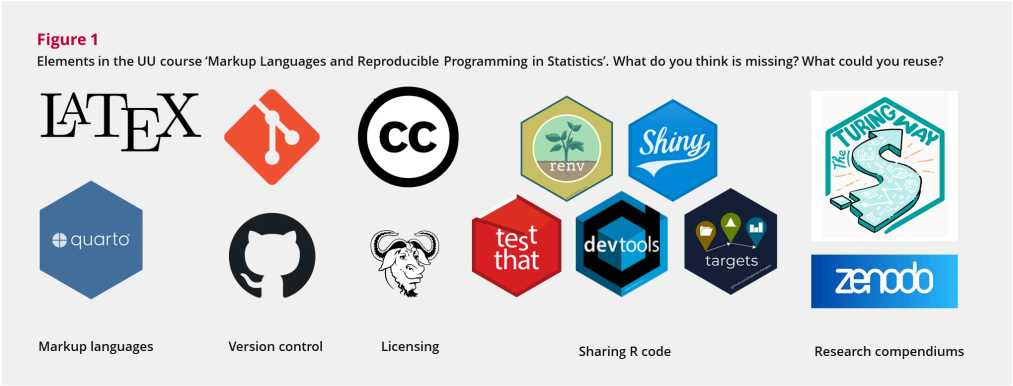
A more domain agnostic resource that we use is **The Turing Way**. The Turing Way is a handbook for reproducible, ethical and collaborative data science, developed by the Turing Way community.³ We also use teaching materials from the research data management support workshops offered by the UU library.^{4,5} But... we might be missing some discipline-independent resources for teaching and testing reproducibility for graduate students.

Missing element(s)?

We do not yet make use of many domain agnostic initiatives for reproducible research. Just to name a few relevant collections: FORRT, Project TIER, and the Carpentries. But more importantly, we could implement **CODECHECK**-ing in our course to teach about computational reproducibility.

A CODECHECK is an attempted reproduction of research outcomes by independent execution of the analysis code. The CODECHECK initiative appoints codecheckers to CODECHECK (peer-reviewed) research articles.⁶

In our course, we could teach students how to do a CODECHECK. We could even use CODECHECK-ing as an assessment tool to evaluate whether students meet the learning objectives. Should we?



References

1. Markup Languages and Reproducible Programming in Statistics team (2024). Course materials. URL: www.gerkovink.com/markup
2. Utrecht University (2024). Course description. URL: <https://osiris-student.uu.nl/#/onderwijs/catalogus/extern/cursus?cursuscode=202000010&taal=en&collegejaar=huidig>
3. The Turing Way Community (2022). The Turing Way: A handbook for reproducible, ethical and collaborative research (1.0.2). DOI: 10.5281/ZENODO.3233853

References

4. Utrecht University (2023). Best Practices for Writing Reproducible Code. URL: utrechtuniversity.github.io/workshop-computational-reproducibility
5. Utrecht University (2023). Writing Reproducible Manuscripts in R & Python. URL: utrechtuniversity.github.io/workshop-reproducible-manuscripts
6. Eglen, S., & Nüst, D., (2024). CODECHECK. URL: codecheck.org.uk



TAKE AWAYS

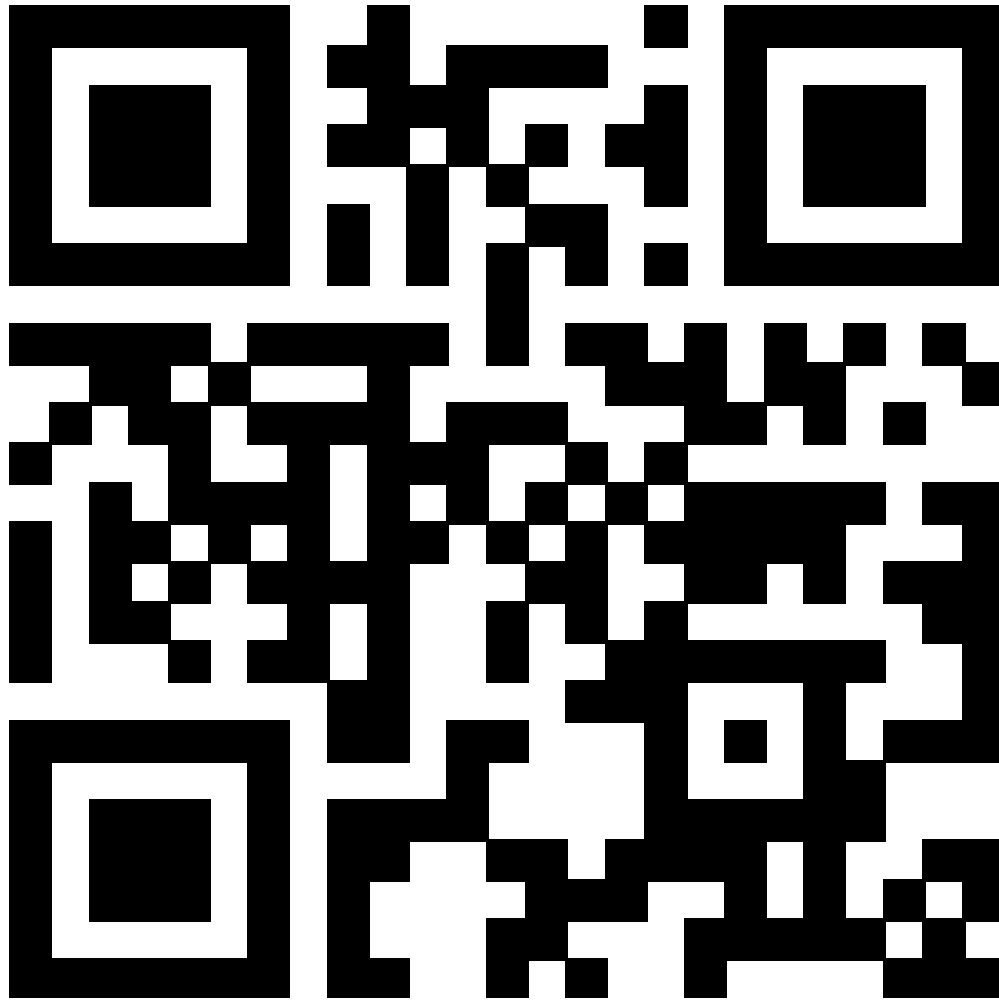
LIKELIHOOD YOU WILL GET CODE WORKING
BASED ON HOW YOU'RE SUPPOSED TO INSTALL IT:



- reproducibility is important
- we should all learn reproducible workflows
- we should teach reproducible workflows



THANK YOU!



edu.nl/nbd3q



RESEARCH COMPENDIUMS



RESEARCH COMPENDIUM



DOI [10.5281/zenodo.8169292](https://doi.org/10.5281/zenodo.8169292)



DEFINITION

A research compendium is a collection of all digital parts of a research project including data, code, texts...

The collection is created in such a way that reproducing all results is straightforward¹

The compendium serves as a means for distributing, managing, and updating the collection²



BASIC COMPENDIUM

A basic research compendium is just a folder...

```
compendium/  
├── data  
│   └── my_data.csv  
├── analysis  
│   └── my_script.R  
├── requirements.txt  
└── README.md
```



(NOT SO) BASIC COMPENDIUM

... but it can become extensive...

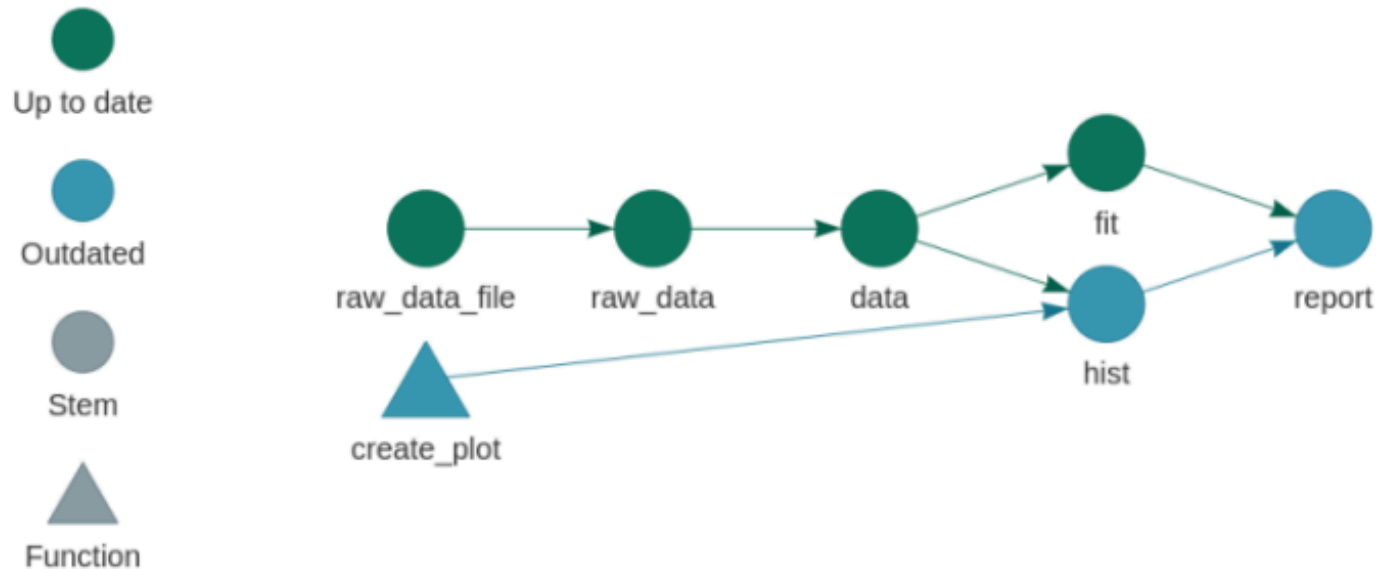
```
|
├── paper/
│   ├── paper.qmd
│   └── references.bib
├── figures/
├── data/
│   ├── raw_data/
│   └── clean_data/
└── templates
    └── journal_template.csl
```



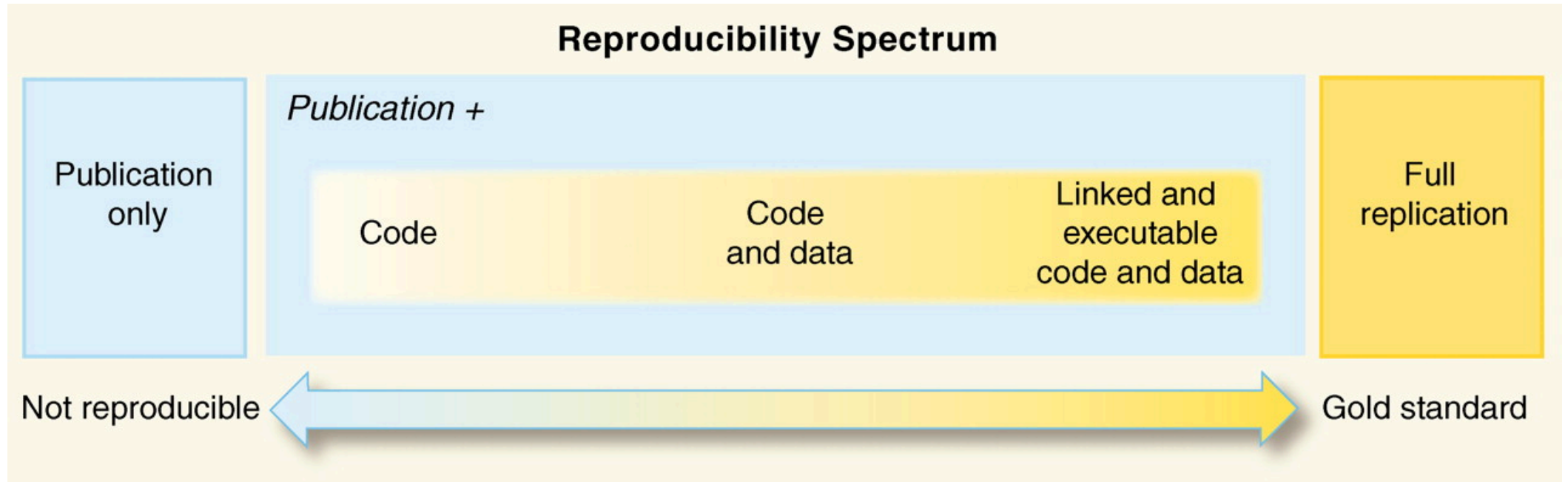
(NOT SO) BASIC COMPENDIUM

...or even executable!

```
|
|— _targets.R
|— R/
|   |— functions_data.R
|   |— functions_analysis.R
|   |— functions_visualization.R
|— data/
|   |— input_data.csv
```



(NOT SO) BASIC COMPENDIUM



GUIDELINES

- Completeness
- Organization
- Economy
- Transparency
- Documentation
- Access
- Provenance
- Metadata
- Automation
- Review



IN PRACTICE



Research Data Management Support workshop:

Writing Reproducible Manuscripts in R and Python



COMPENDIUM STEP-BY-STEP

- Think about a good folder structure
 - Split up 'read-only', 'human-generated', and 'project-generated' files
- Create folder structure (main directory and sub directories)
 - Add a landing page in the form of a README document
 - Make the compendium executable (to automatically generate the results; optional)
 - Make the compendium into a git repository (optional)
- Add all files needed for reproducing the results of the project
 - Avoid 'hard coded' parameters or human intervention in the execution
- Make the compendium as clean and easy to use as possible
 - Include a citation file and a LICENSE file with info on how it can be used
- Publish your compendium
 - E.g. on Zenodo (optional, more on this in the last course week)



REFERENCES

Markup Languages and Reproducible Programming in Statistics team (2024). Course materials. URL: www.gerkovink.com/markup

Utrecht University (2024). Course description. URL: <https://osiris-student.uu.nl/#/onderwijscatalogus/extern/cursus?cursuscode=202000010&taal=en&collegejaar=huidig>

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